KAIZEN MATHS

Lesson starter

Trigonometric Differentiation

Mixed Trig Challenges

Name _____

S1

Differentiate with respect to x.

$$y = \frac{\sec(-2x+3)}{\cos(-7x-8)}$$

S2

Differentiate with respect to x.

$$y = (\sec(2x-1))^2 \cot(2x-7)$$

S3

Differentiate with respect to x.

$$y = \csc(8x+5)\sin(-9x-2)$$

S4

Differentiate with respect to x.

$$y = \frac{\sec(7x-8)}{\sin(3x-1)}$$

Show your working

KAIZEN MATHS

Lesson starter

Trigonometric Differentiation

Mixed Trig Challenges

Name _____

S1

Differentiate with respect to x.

$$y = \frac{\sec(-2x+3)}{\cos(-7x-8)}$$

S2

Differentiate with respect to x.

$$y = (\sec(2x-1))^2 \cot(2x-7)$$

S3

Differentiate with respect to x.

$$y = \csc(8x+5)\sin(-9x-2)$$

S4

Differentiate with respect to x.

$$y = \frac{\sec(7x-8)}{\sin(3x-1)}$$

ALG-C0E7-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Trigonometric Differentiation

Mixed Trig Challenges

Name _____

E1

Differentiate with respect to x.

$$y = \frac{\sin(-9x - 1)}{\sec(5x + 9)}$$

E2

Differentiate with respect to x.

$$y = (\sin(4x^2 - 5x - 2))^4$$

E3

Differentiate with respect to x.

$$y = \frac{\csc(6x + 5)}{\sin(7x - 2)}$$

Show your working

KAIZEN MATHS

Exit ticket

Trigonometric Differentiation

Mixed Trig Challenges

Name _____

E1

Differentiate with respect to x.

$$y = \frac{\sin(-9x - 1)}{\sec(5x + 9)}$$

E2

Differentiate with respect to x.

$$y = (\sin(4x^2 - 5x - 2))^4$$

E3

Differentiate with respect to x.

$$y = \frac{\csc(6x + 5)}{\sin(7x - 2)}$$

ALG-C0E7-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Trigonometric Differentiation

Mixed Trig Challenges

Name _____

E1

Differentiate with respect to x.

$$y = \frac{\sin(-9x - 1)}{\sec(5x + 9)}$$

E2

Differentiate with respect to x.

$$y = (\sin(4x^2 - 5x - 2))^4$$

E3

Differentiate with respect to x.

$$y = \frac{\csc(6x + 5)}{\sin(7x - 2)}$$

ALG-C0E7-L4 · E

Show your working

KAIZEN MATHS

Exit ticket

Trigonometric Differentiation

Mixed Trig Challenges

Name _____

E1

Differentiate with respect to x.

$$y = \frac{\sin(-9x - 1)}{\sec(5x + 9)}$$

E2

Differentiate with respect to x.

$$y = (\sin(4x^2 - 5x - 2))^4$$

E3

Differentiate with respect to x.

$$y = \frac{\csc(6x + 5)}{\sin(7x - 2)}$$

ALG-C0E7-L4 · E

Show your working

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

ALG-C0E7-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Differentiate with respect to x.

$$y = \frac{\sec(-2x+3)}{\cos(-7x-8)}$$

Answer:

$$\frac{dy}{dx} = \frac{-\sec(-2x+3)[2\tan(-2x+3)\cos(-7x-8)+7\sin(-7x-8)]}{(\cos(-7x-8))^2}$$

Working:

Identify the numerator and denominator: $u = \sec(-2x+3)$ and $v = \cos(-7x-8)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (\sec(-2x+3)\tan(-2x+3))(-2)$

$u' = -2\sec(-2x+3)\tan(-2x+3)$

$v' = (-\sin(-7x-8))(-7)$

$v' = 7\sin(-7x-8)$

Substitute into the quotient rule before simplifying.

$$\frac{dy}{dx} = \frac{(-2\sec(-2x+3)\tan(-2x+3))\cos(-7x-8) - \sec(-2x+3)(7\sin(-7x-8))}{(\cos(-7x-8))^2}$$

Factor out the common factor $-\sec(-2x+3)$.

Factorised final form: $\frac{dy}{dx} = \frac{-\sec(-2x+3)[2\tan(-2x+3)\cos(-7x-8)+7\sin(-7x-8)]}{(\cos(-7x-8))^2}$

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

ALG-C0E7-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S2

Differentiate with respect to x.

$y = (\sec(2x-1))^2 \cot(2x-7)$

Answer:

$\frac{dy}{dx} = 2(\sec(2x-1))^2[2\tan(2x-1)\cot(2x-7) - (\csc(2x-7))^2]$

Working:

This is a product, and the first factor also needs the chain rule.

Let $f = (\sec(2x-1))^2$ and $g = \cot(2x-7)$.

$f' = 2(\sec(2x-1))^{2-1}(\sec(2x-1)\tan(2x-1))(2)$

$f' = 4(\sec(2x-1))^1 \sec(2x-1)\tan(2x-1)$

$g' = (-\csc^2(2x-7))(2)$

$g' = -2\csc^2(2x-7)$

Apply product rule: $y' = f'g + fg'$.

$y' = (4(\sec(2x-1))^1 \sec(2x-1)\tan(2x-1))(\cot(2x-7)) + ((\sec(2x-1))^2)(-2\csc^2(2x-7))$

$y' = 4(\sec(2x-1))^1 \sec(2x-1)\tan(2x-1)\cot(2x-7) - 2(\sec(2x-1))^2 \csc^2(2x-7)$

Factor out the common factor $2(\sec(2x-1))^2$.

Factorised final form: $\frac{dy}{dx} = 2(\sec(2x-1))^2[2\tan(2x-1)\cot(2x-7) - (\csc(2x-7))^2]$

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

ALG-C0E7-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Differentiate with respect to x.

$y = \csc(8x + 5)\ \sin(-9x - 2)$

Answer:

$\frac{dy}{dx} = -\csc(8x + 5)[8\ \cot(8x + 5)\ \sin(-9x - 2) + 9\ \cos(-9x - 2)]$

Working:

Write the two factors as $f = \csc(8x + 5)$ and $g = \sin(-9x - 2)$.

Use product rule: $(fg)' = f'g + fg'$.

$f' = (-\csc(8x + 5)\cot(8x + 5))(8)$

$f' = -8\ \csc(8x + 5)\cot(8x + 5)$

$g' = (\cos(-9x - 2))(-9)$

$g' = -9\ \cos(-9x - 2)$

Substitute into the product rule: $y' = (-8\ \csc(8x + 5)\cot(8x + 5))(\sin(-9x - 2)) + (\csc(8x + 5))(-9\ \cos(-9x - 2))$

.

$y' = -8\ \csc(8x + 5)\cot(8x + 5)\ \sin(-9x - 2) - 9\ \csc(8x + 5)\ \cos(-9x - 2)$

Factor out the common factor $-\csc(8x + 5)$.

Factorised final form: $\frac{dy}{dx} = -\csc(8x + 5)[8\ \cot(8x + 5)\ \sin(-9x - 2) + 9\ \cos(-9x - 2)]$

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

ALG-C0E7-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

Differentiate with respect to x.

$$y = \frac{\sec(7x - 8)}{\sin(3x - 1)}$$

Answer:

$$\frac{dy}{dx} = \frac{\sec(7x - 8)[7\ \tan(7x - 8)\ \sin(3x - 1) - 3\ \cos(3x - 1)]}{(\sin(3x - 1))^2}$$

Working:

Identify the numerator and denominator: $u = \sec(7x - 8)$ and $v = \sin(3x - 1)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (\sec(7x - 8)\tan(7x - 8))(7)$

$u' = 7\ \sec(7x - 8)\tan(7x - 8)$

$v' = (\cos(3x - 1))(3)$

$v' = 3\ \cos(3x - 1)$

Substitute into the quotient rule before simplifying.

$$\frac{dy}{dx} = \frac{(7\ \sec(7x - 8)\tan(7x - 8))\ \sin(3x - 1) - \sec(7x - 8)\ (3\ \cos(3x - 1))}{(\sin(3x - 1))^2}$$

Factor out the common factor $\sec(7x - 8)$.

Factorised final form: $\frac{dy}{dx} = \frac{\sec(7x - 8)[7\ \tan(7x - 8)\ \sin(3x - 1) - 3\ \cos(3x - 1)]}{(\sin(3x - 1))^2}$

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

ALG-C0E7-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Differentiate with respect to x.

$$y = \frac{\sin(-9x - 1)}{\sec(5x + 9)}$$

Answer:

$$\frac{dy}{dx} = \frac{-[9\cos(-9x - 1) + 5\sin(-9x - 1)\tan(5x + 9)]}{\sec(5x + 9)}$$

Working:

Identify the numerator and denominator: $u = \sin(-9x - 1)$ and $v = \sec(5x + 9)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (\cos(-9x - 1))(-9)$

$u' = -9\cos(-9x - 1)$

$v' = (\sec(5x + 9)\tan(5x + 9))(5)$

$v' = 5\sec(5x + 9)\tan(5x + 9)$

Substitute into the quotient rule before simplifying.

$$\frac{dy}{dx} = \frac{(-9\cos(-9x - 1))\sec(5x + 9) - \sin(-9x - 1)(5\sec(5x + 9)\tan(5x + 9))}{(\sec(5x + 9))^2}$$

Factor out the common factor $-\sec(5x + 9)$.

$$y' = \frac{-\sec(5x + 9)[9\cos(-9x - 1) + 5\sin(-9x - 1)\tan(5x + 9)]}{(\sec(5x + 9))^2}$$

Cancel common numerator and denominator factors.

Factorised final form: $\frac{dy}{dx} = \frac{-[9\cos(-9x - 1) + 5\sin(-9x - 1)\tan(5x + 9)]}{\sec(5x + 9)}$

E2

Differentiate with respect to x.

$$y = (\sin(4x^2 - 5x - 2))^4$$

Answer:

$$\frac{dy}{dx} = 4(\sin(4x^2 - 5x - 2))^3\cos(4x^2 - 5x - 2)(8x - 5)$$

Working:

Work from the outside in: power, then trig function, then polynomial inner function.

Outer power derivative gives $4(\sin(4x^2 - 5x - 2))^3$.

Trig derivative gives $\cos(4x^2 - 5x - 2)$.

Inner derivative gives $8x - 5$.

Multiply the three derivative parts together.

$$\frac{dy}{dx} = 4(\sin(4x^2 - 5x - 2))^3\cos(4x^2 - 5x - 2)(8x - 5)$$

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

ALG-C0E7-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Differentiate with respect to x.

$$y = \frac{\csc(6x+5)}{\sin(7x-2)}$$

Answer:

$$\frac{dy}{dx} = \frac{-\csc(6x+5)[6\cot(6x+5)\sin(7x-2)+7\cos(7x-2)]}{(\sin(7x-2))^2}$$

Working:

Identify the numerator and denominator: $u = \csc(6x+5)$ and $v = \sin(7x-2)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (-\csc(6x+5)\cot(6x+5))(6)$

$u' = -6\csc(6x+5)\cot(6x+5)$

$v' = (\cos(7x-2))(7)$

$v' = 7\cos(7x-2)$

Substitute into the quotient rule before simplifying.

$$\frac{dy}{dx} = \frac{(-6\csc(6x+5)\cot(6x+5))\sin(7x-2) - \csc(6x+5)(7\cos(7x-2))}{(\sin(7x-2))^2}$$

Factor out the common factor $-\csc(6x+5)$.

Factorised final form: $\frac{dy}{dx} = \frac{-\csc(6x+5)[6\cot(6x+5)\sin(7x-2)+7\cos(7x-2)]}{(\sin(7x-2))^2}$